

Patients on long-term oxygen therapy must haul a portable concentrator everywhere — stairs are the highest fall-risk, highest-exertion obstacle in the home.



- Older stair-users show higher stride variability & fall risk (**Startzell 2000; Hausdorff 1997**).
- Our robot **carries the ~2.2 kg concentrator**, follows autonomously, detects stairs, and climbs — beside the patient.
- Both hands freed for the railing; robot handles the load.

Robot: Unitree Go2, 6.9 kg, 12-DOF
Compute: NVIDIA Jetson Orin
Vision: Intel RealSense D435 RGB-D
LiDAR: Hesai XT16
Payload: 2.22 kg mock O₂ + cradle
 (~32 % of trunk mass)

Fig. 2 — Go2 with payload cradle

System overview — stair-climbing robotic dog · live [run_sim.bat](#) path

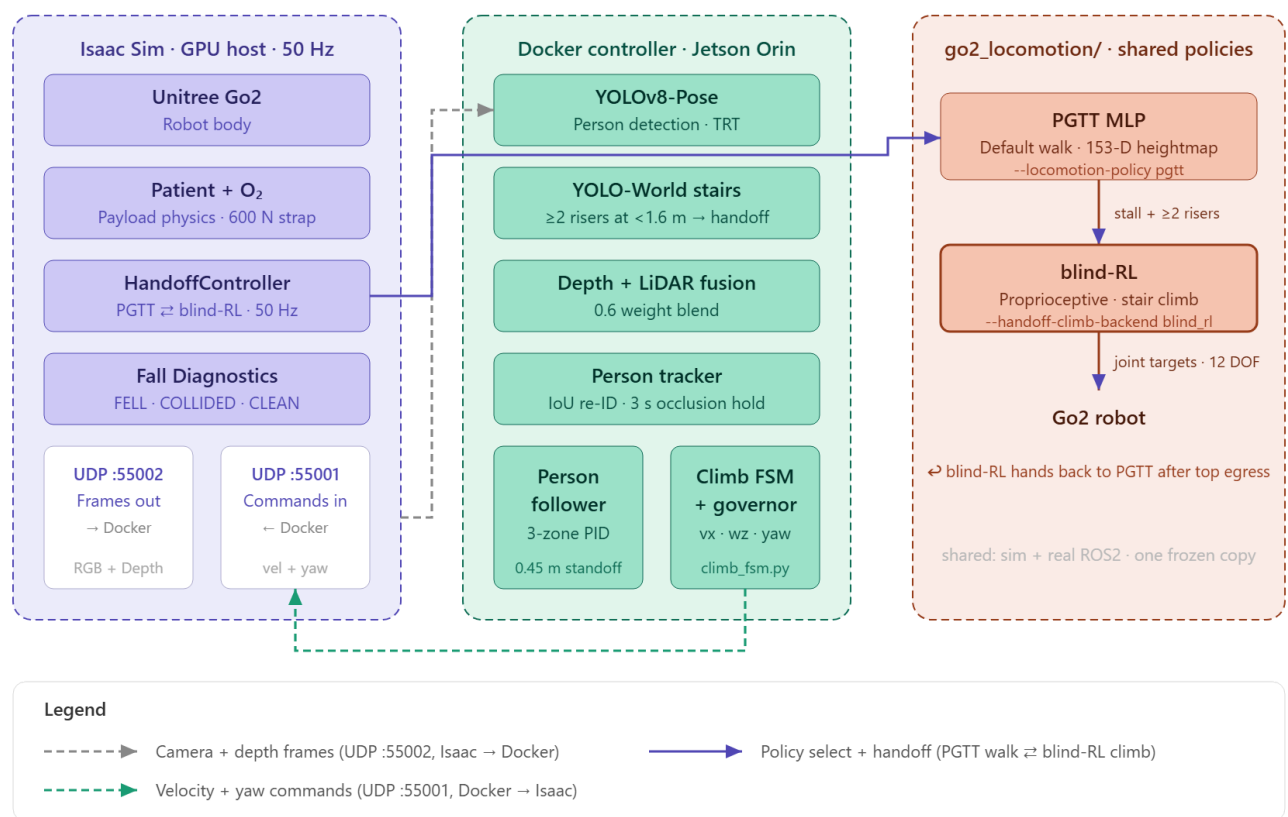


Fig. 3 — One shared codebase runs in NVIDIA Isaac Sim and on the real Go2; only the sensor source is swapped. Perception + control (Jetson / Docker) emit velocity commands; a shared policy library returns 12-DOF joint targets.

- 1 **Perceive** — RealSense D435 RGB-D + LiDAR feed two parallel vision heads: YOLOv8-Pose (person detection) and YOLO-World (open-vocab stair detection).
- 2 **Follow** — ByteTrack lock with 3 s occlusion hold. 3-zone PID holds 0.45 m standoff; depth-reactive avoidance routes around furniture.
- 3 **Detect Stairs** — YOLO-World counts risers; LiDAR depth corroborates (0.6 weight blend). ≥ 2 risers within 1.6 m triggers handoff. Dual-evidence gate eliminates false positives.
- 4 **Hand Off** — 50 Hz HandoffController (WALK $\leftrightarrow$ CLIMB $\leftrightarrow$ TOP-EGRESS / ABORT) swaps PGTT walker for blind-RL climb policy on motion-still + riser trigger.
- 5 **Climb** — Proprioceptive “blind” RL policy drives legs up; at crest the robot walks clear and returns to walk policy.

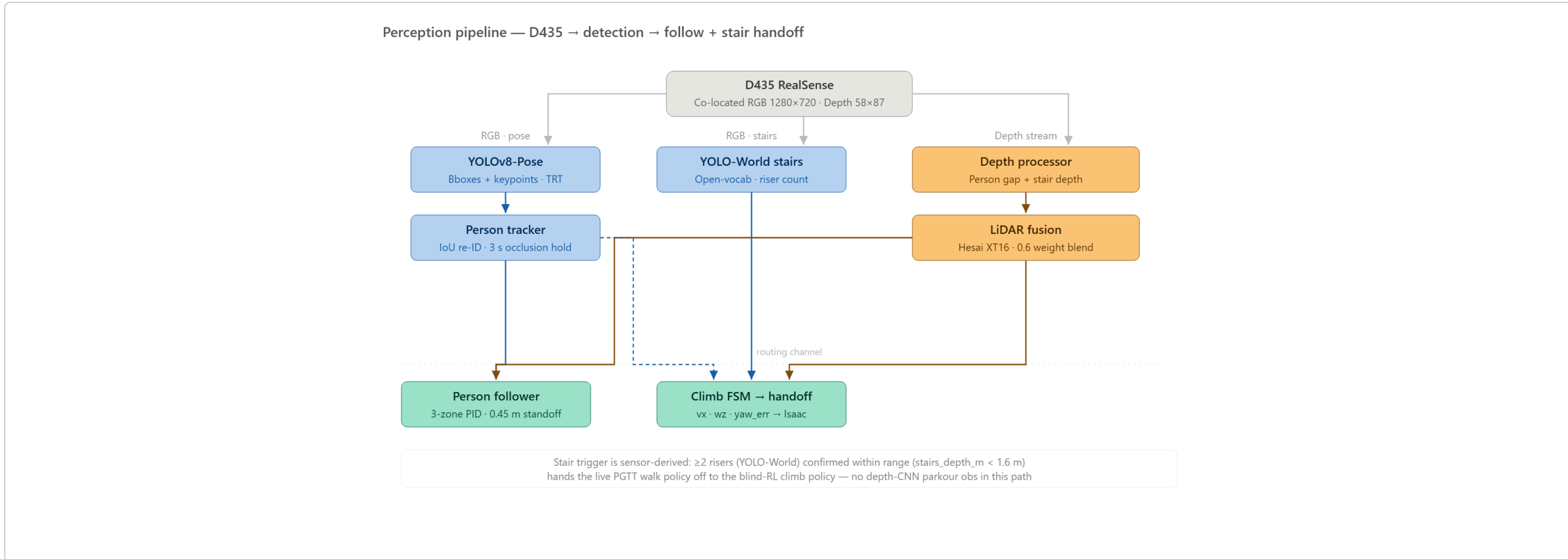


Fig. 4 — Perception pipeline: sensor inputs feed parallel vision heads; outputs merge in the Climb FSM

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Full climbs
(0.12–0.15 m risers)

0.15 m

ADA-max riser
successfully topped

≤31°

Peak body tilt
(threshold: 60°)

How far up the stairs (steps climbed)

Riser Height	Steps Climbed
0.12m ~50°	14.0
0.13m ~50°	13.0
0.14m ~50°	13.0
0.15m 6° ADA maximum	14.0
0.17m 7° IRC standard	4.4
0.19m 7.75° comm. max	0.0

Stability: peak tilt vs fall threshold

Riser Height	Max Body Tilt (deg)
0.12m ~50°	28
0.13m ~50°	26
0.14m ~50°	27
0.15m 6° ADA maximum	31
0.17m 7° IRC standard	32
0.19m 7.75° comm. max	6

Fig. 5 — Steps climbed (of 14) vs. riser height

Fig. 6 — Peak body tilt stays well below 60° fall threshold.

Riser	Steps (/ 14)	Peak Tilt	Outcome
0.120 m (~5°)	14.0	28°	✓ Full climb
0.130 m (~5°)	13.9	24°	✓ Full climb
0.140 m (~6°)	13.9	27°	✓ Full climb
0.150 m (6° ADA)	14.0	31°	✓ Full climb
0.175 m (7° IBC)	4.4	32°	✗ Partial
0.198 m (7.75°)	0.0	6°	✗ No climb

Robot topped the ADA-maximum (0.15 m) staircase end-to-end in simulation, staying upright throughout (peak tilt $\leq 31^\circ$, well below the 60° fall threshold).

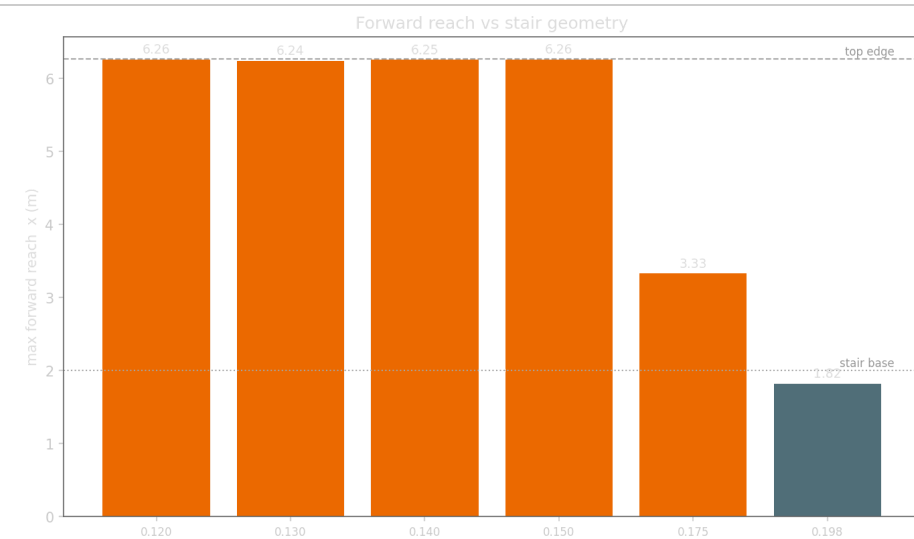


Fig. 7 — Max forward reach across the staircase at each riser height; robot reaches the top edge for 0.12–0.15 m risers.

About 1.5 million Americans rely on long-term home oxygen therapy, and stairs are consistently the task they fear most — a fall risk that can confine a patient to one floor of their own home. This project builds a sensor-guided quadruped that walks beside the patient, carries the oxygen equipment, and climbs the stairs on its own — so the patient never has to face a staircase alone.

The diagram illustrates two state machines for a robot's movement.

Left State Machine: HandoffController — Isaac Sim 50 Hz

- WALK** (blue box): PGTG MLP - heightmap walk.
 - Transitions:
 - reset = walk** (dashed arrow) back to WALK.
 - stall = riser > 2** (dashed arrow) to CLIMB.
- CLIMB** (blue box): blind-RL, proprioceptive climb.
 - Transitions:
 - crest clear** (dashed arrow) to TOP EGRESS.
 - ABORT?** (solid arrow, labeled "Timeout / RL") back to WALK.
- TOP EGRESS** (blue box): Walk clear of stair crest.
 - Transitions:
 - walk resumed** (dashed arrow) back to WALK.

Right State Machine: ClimbFSM — Docker climb_fsm.py

- FLAT FOLLOW** (green box): Normal tracking.
 - Transitions:
 - person lost** (dashed arrow) to LOSS OF FLOOR Recovery.
 - stairs detected** (solid arrow) to STAIR APPROACH.
- STAIR APPROACH** (green box): Slowing - aligning.
 - Transitions:
 - range close** (solid arrow) to STAIR NEAR.
- STAIR NEAR** (green box): Handoff pending.
 - Transitions:
 - handoff active** (solid arrow) to COMMITTED CLIMB.
- COMMITTED CLIMB** (green box): Handoff active - blind-RL.
 - Transitions:
 - stairs gone** (solid arrow) to LOSS OF FLOOR Recovery.
- LOSS OF FLOOR Recovery** (orange box):
 - Transitions:
 - person required — flat follow** (dashed arrow) back to FLAT FOLLOW.
 - timeout** (solid arrow) to STOP.
- STOP** (grey box): Await reacquire.

Fig. 8 — HandoffController (Isaac Sim, 50 Hz) and ClimbFSM (Docker) run concurrently via UDP

Fig. 10 — Real learning curve: 6,000 PPO iterations, mean reward converges to

Fig. 9 — Follow mode: robot trails the patient at a 0.45 m standoff, carrying the O₂ tank.

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blind-RL climb policy — proprioceptive rl_sar MLP · how it works & was built

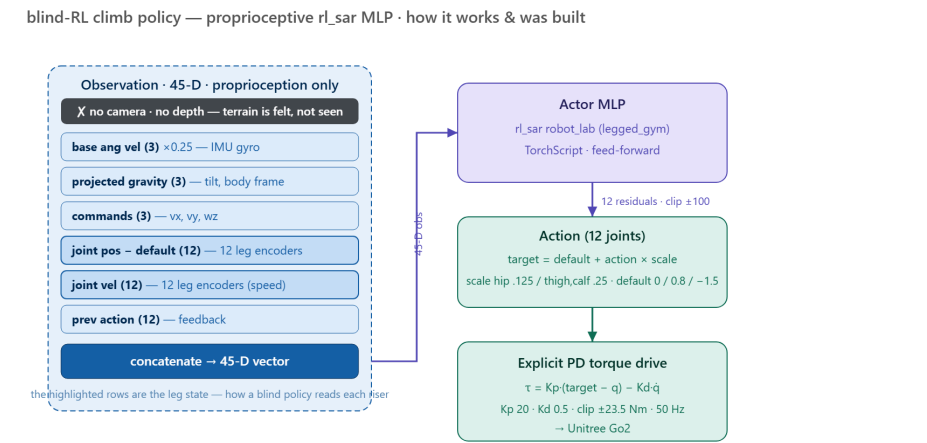


Fig. 11 — Blind-RL policy: 45-D proprioceptive input $\rightarrow$ Actor MLP $\rightarrow$ 12-DOF joint targets at 50 Hz. No camera, no depth — terrain is felt via leg load + IMU

- End-to-end **follow** → **detect** → **hand-off** → **climb** stack validated in NVIDIA Isaac Sim on a shared sim/real codebase (**401 passing tests**).
- Successfully reaches the top of the **0.15 m (ADA-maximum)** staircase end-to-end in the best demo.
- Peak tilt $\leq 31^\circ$ across all successful runs — **well below the 60° fall threshold**.

- **Nose-down gait** — blind- κ L climber pitches forward; a gait-posture problem, not a handoff bug.
- **High CoG** — O_2 carrier raises center of gravity; wider stance being explored.
- **Safety grade-gate** — stance locking suppressed mid-climb; min observed clearance 0.209 m vs. 0.55 m target.
- **Hardware climb unproven** — physical Go2 validation is the active next step.

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 [3] Zhuang et al. (2025). *PGTT*. arXiv:2510.18348. [4] Wang et al. (2023). *YOLO-world*. arXiv:2401.17270.
 [5] Zhang et al. (2022). *ByteTrack*. ECCV 2022. [6] NVIDIA Isaac Sim - Unittree Go2 EDU - Intel RealSense D435.